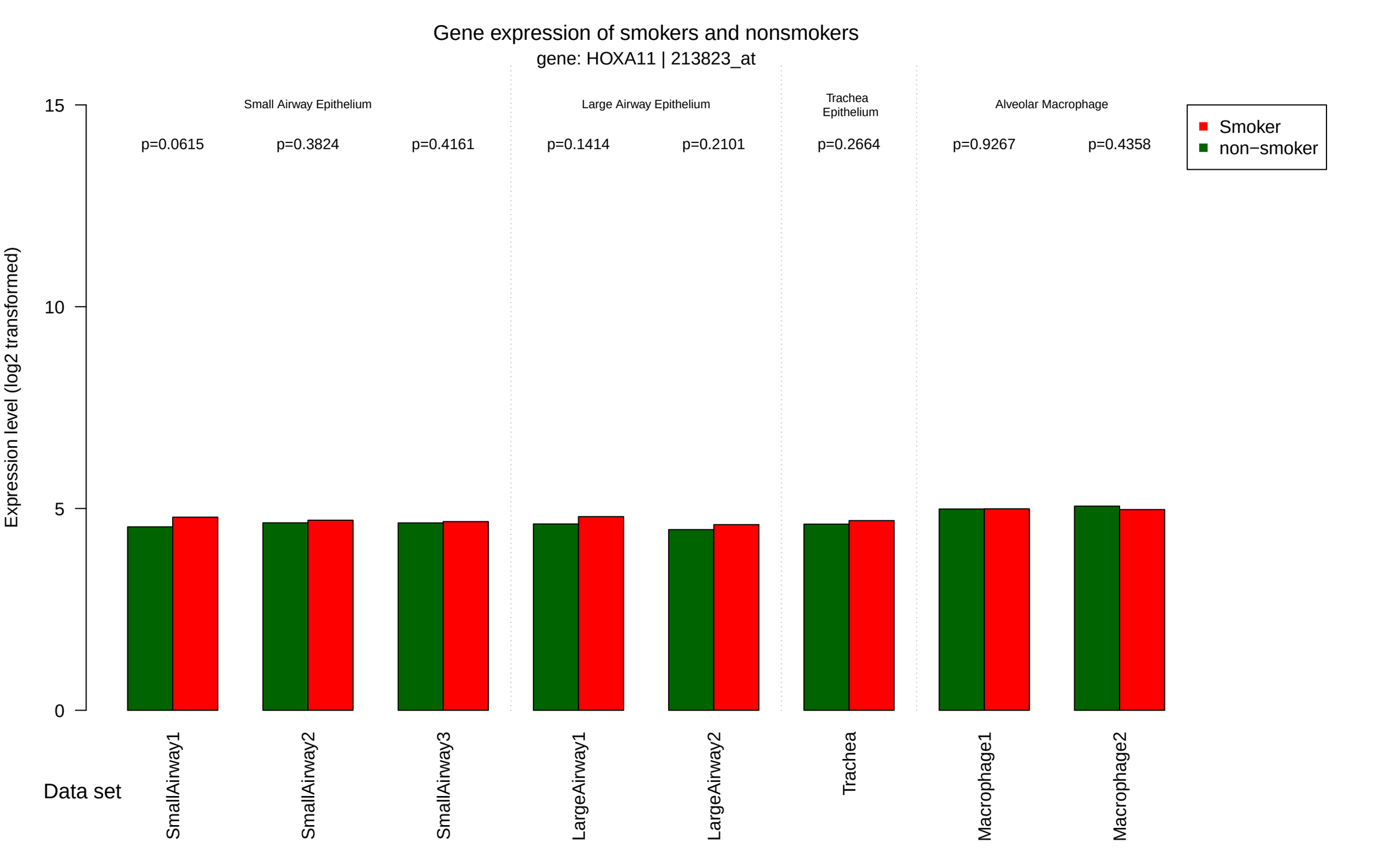
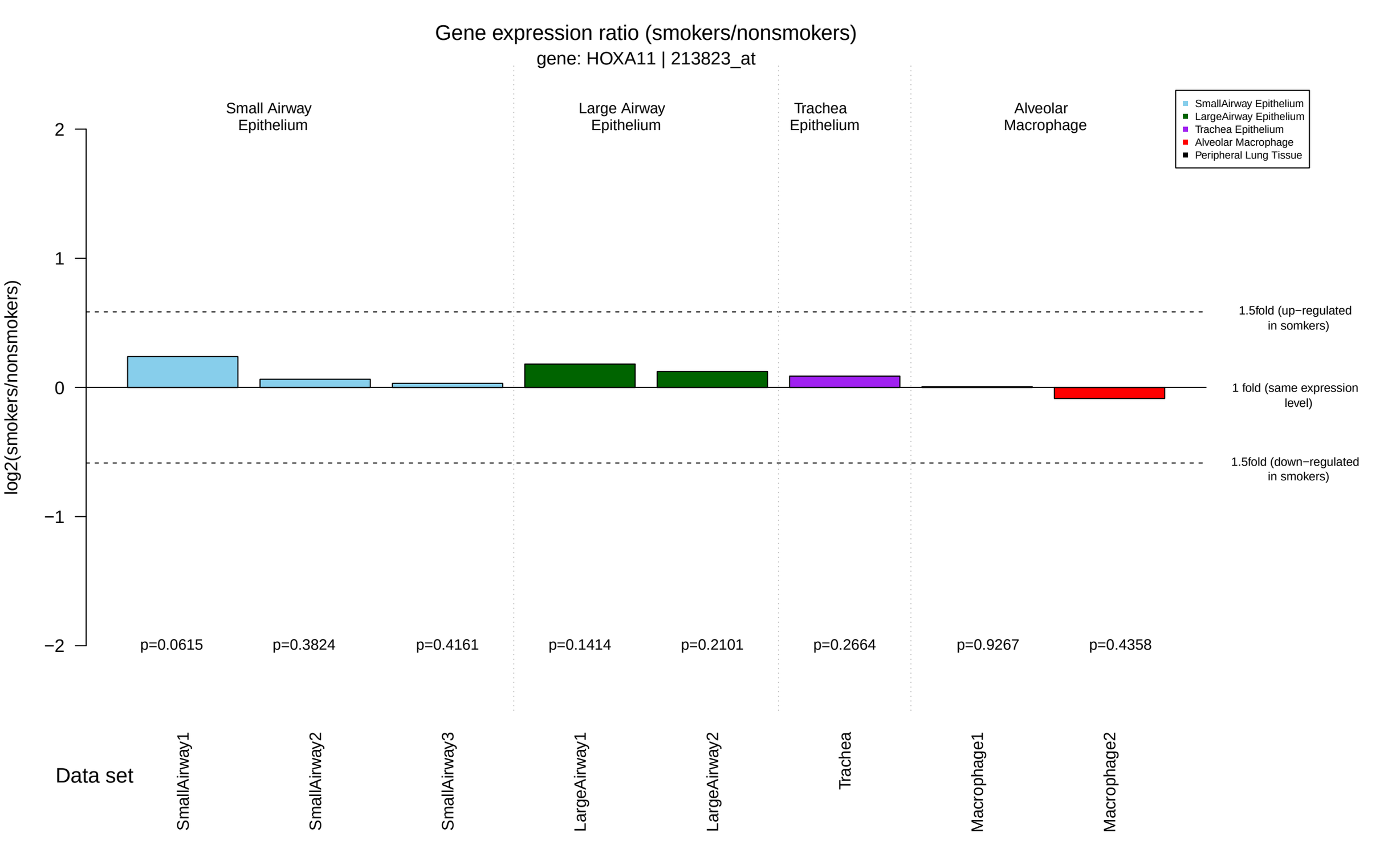
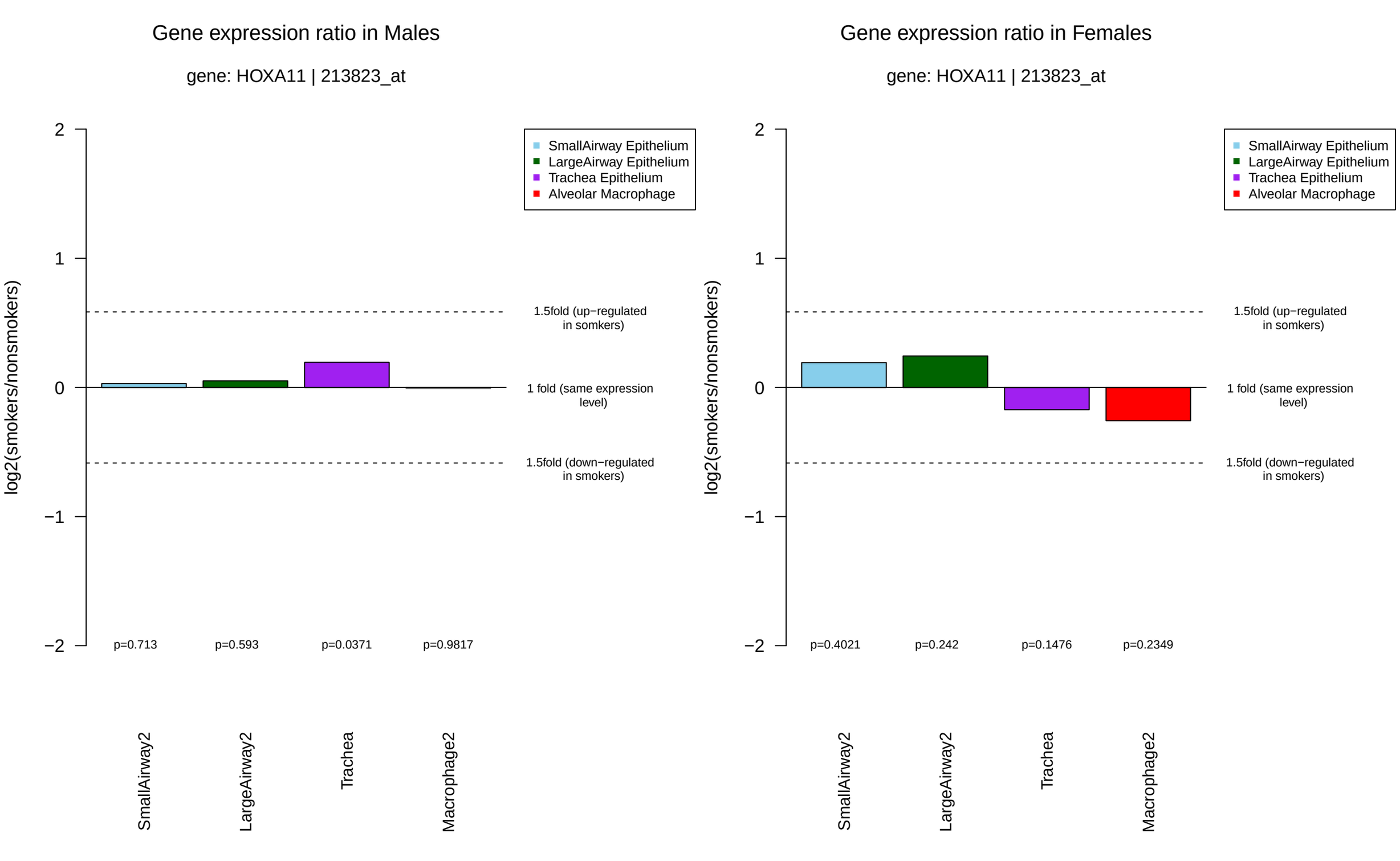
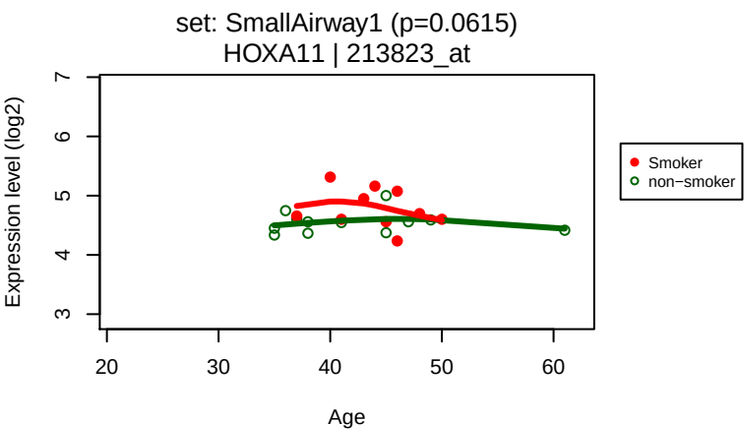
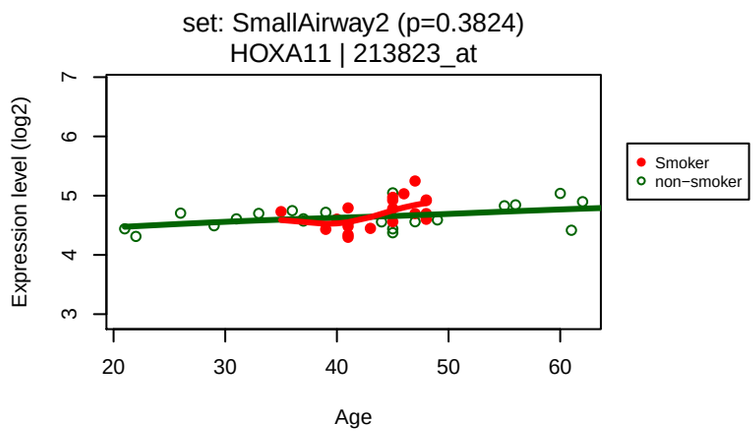
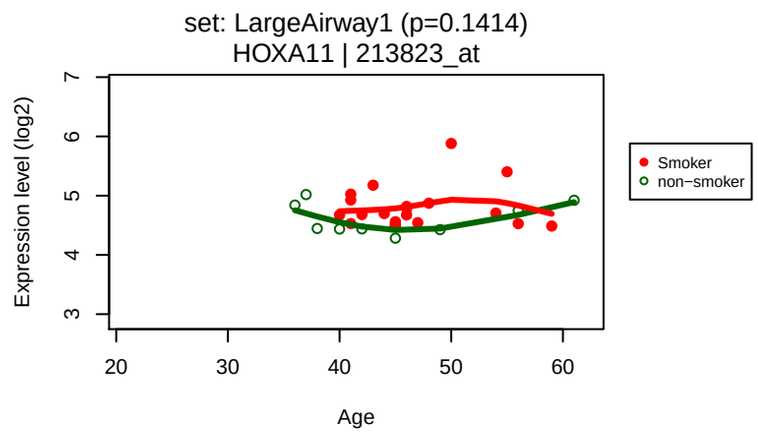

Gene expression of smokers and nonsmokers
gene: HOXA11 | 213823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0615
p=0.3824
p=0.4161
p=0.1414
p=0.2101
p=0.2664
p=0.9267
p=0.4358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

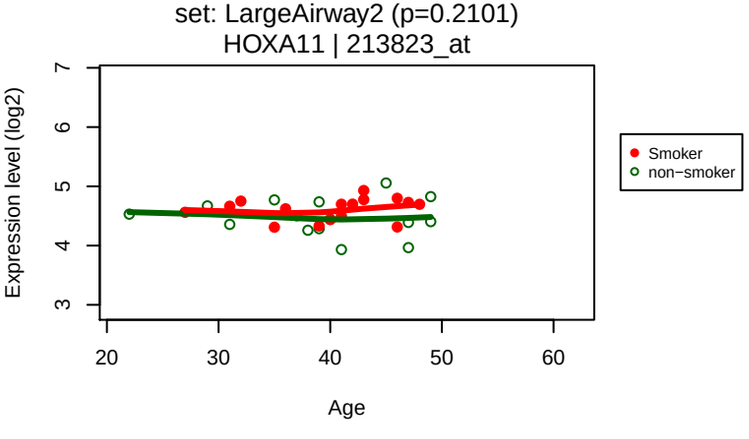

Gene expression ratio (smokers/nonsmokers)
gene: HOXA11 | 213823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0615
p=0.3824
p=0.4161
p=0.1414
p=0.2101
p=0.2664
p=0.9267
p=0.4358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

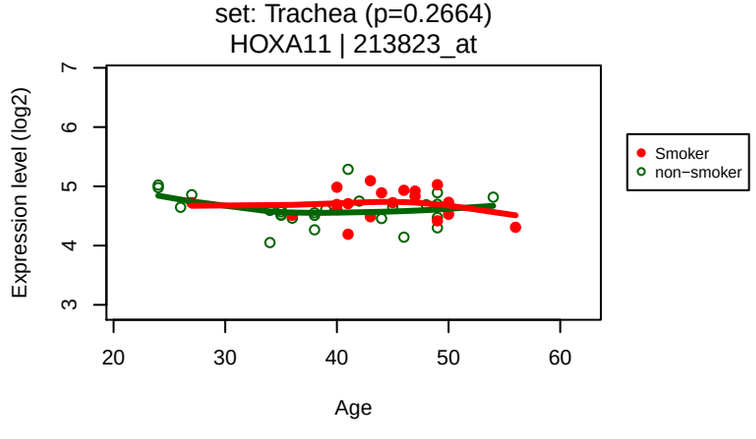

Gene expression ratio in Males
gene: HOXA11 | 213823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.713
p=0.593
p=0.0371
p=0.9817
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA11 | 213823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4021
p=0.242
p=0.1476
p=0.2349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

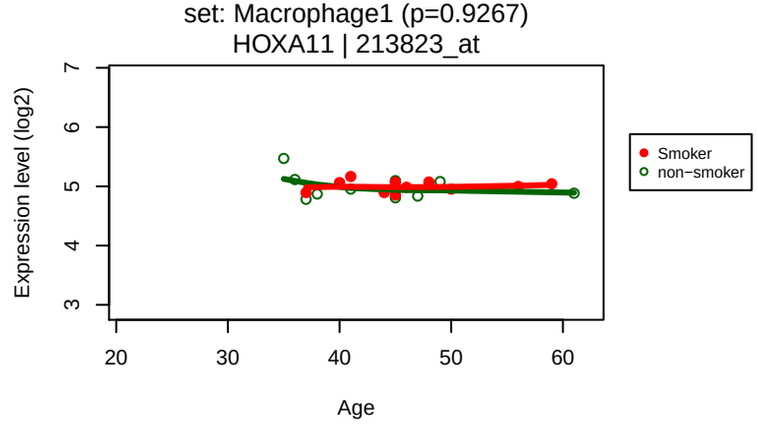

set: SmallAirway1 (p=0.0615)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker

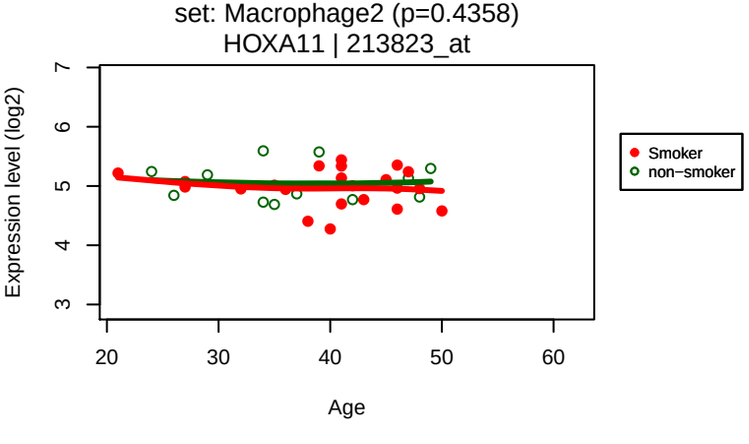

set: SmallAirway2 (p=0.3824)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker

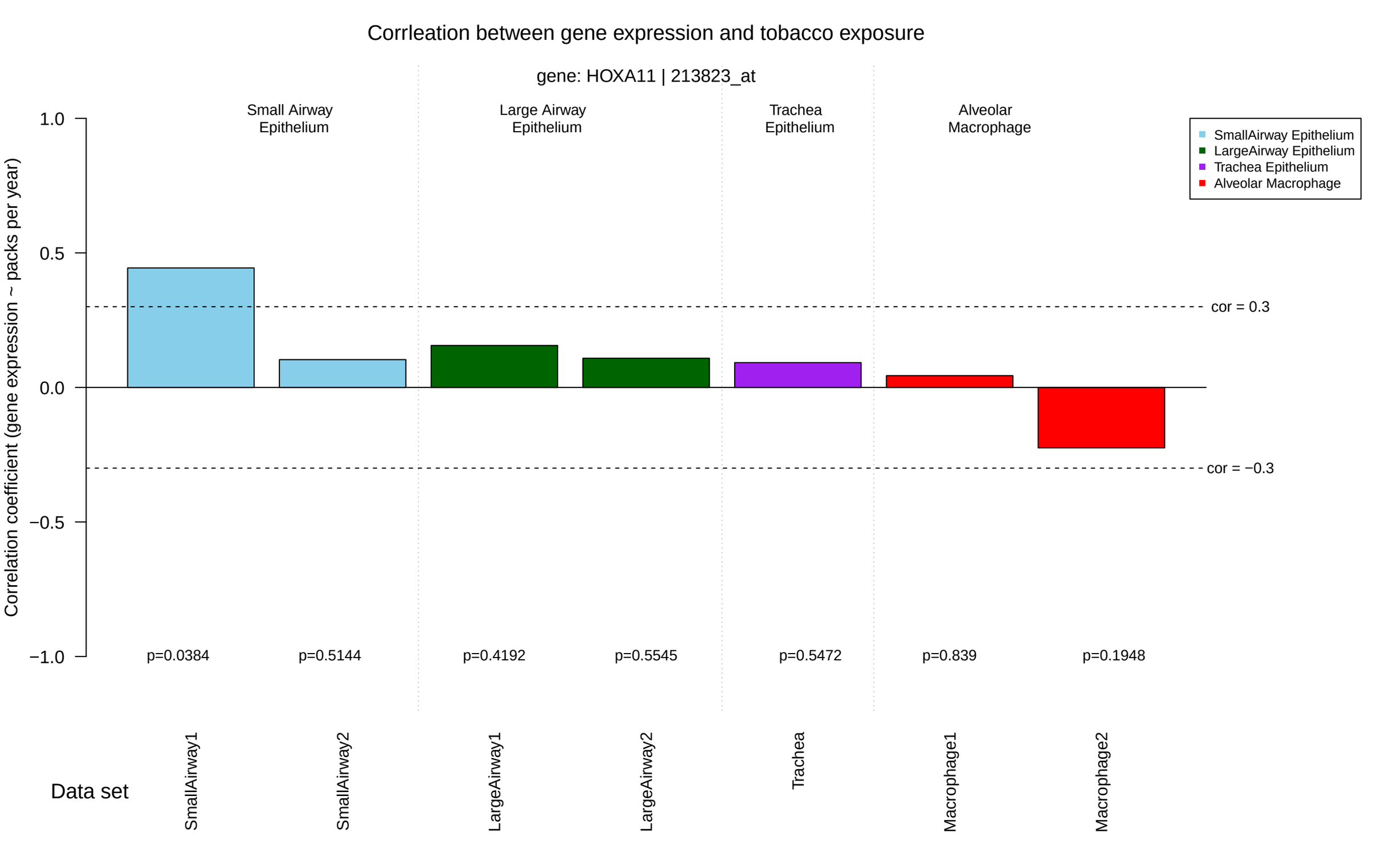

set: LargeAirway1 (p=0.1414)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2101)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2664)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9267)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4358)
HOXA11 | 213823_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA11 | 213823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0384
p=0.5144
p=0.4192
p=0.5545
p=0.5472
p=0.839
p=0.1948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2